GEOMETRY · UNIT 9: CIRCLES

Pizza Parlor Geometry

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students analyze a pizza menu with circle geometry. They compare areas and price per square inch, find crust (arc) lengths and slice (sector) areas, use inscribed angles, chords, tangents, and circle equations, and design a menu with fair prices.

Area grows with the square of the radius, which surprises students when they compare pizza sizes. Arcs and sectors become crust and slices, and circle theorems appear in cutting and boxing.

Standards Addressed

- G-C.2 Relationships among inscribed angles, radii, and chords
- G-C.5 Arc length and sector area
- G-GMD.1 Circumference and area of a circle
- G-GPE.1 Equation of a circle (CP1 and Honors)

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: one large or two mediums? Vote. Part 1 area and value.
Day 2	Part 2 arcs, sectors, and circle theorems.
Day 3	Part 3 menu design.

Materials

- Student handout
- Calculators
- Paper plates and protractors (optional)
- Real pizza menu (optional)

How the Three Versions Differ

Level	What changes
CP2	Area, circumference, price per square inch, and slices as fractions of the circle.
CP1	Sector angle from area, chord length, inscribed angles, and a delivery-zone equation.
Honors	Segment area, intersecting chords and secants, and converting a delivery zone from general form.

Watch For (Common Misconceptions)

- Using diameter instead of radius in πr^2.
- Thinking doubling the diameter doubles the area.
- Confusing arc measure (degrees) with arc length (inches).

Extension Ideas

- Survey local pizzerias' prices.
- Design a square pizza with the same area.
- Explore why pizzas are round.

Answer Key – CP2

Part 1: Which Pizza Is the Better Deal?

#	Task	Answer
1	Find the area of a large (14-inch) pizza. Give an exact answer in terms of π and a decimal.	$\pi(7)^2 = 49\pi \approx 153.94$ in²
2	Find the area of one medium (10-inch) pizza and of two mediums. Which is more pizza: one large or two mediums?	$25\pi \approx 78.54$ in²; two mediums $50\pi \approx 157.08$ in². Two mediums are slightly more pizza.
3	Find the price per square inch for the large and for two mediums. Which is the better deal?	Large: \$15 ÷ 153.94 ≈ \$0.097 per in². Two mediums: \$18 ÷ 157.08 ≈ \$0.115 per in². The large is the better deal.
4	How long is the crust (circumference) of a large pizza?	$14\pi \approx 43.98$ in

Part 2: Slices

#	Task	Answer
5	What is the central angle of each slice?	360° ÷ 8 = 45°
6	Find the crust length (arc length) of one slice.	$(45 \div 360) \times 14\pi = 1.75\pi \approx 5.50$ in
7	Find the area of one slice (sector area).	$(45 \div 360) \times 49\pi = 6.125\pi \approx 19.24$ in²
8	A 12-inch pizza is cut into 6 slices. Is one of its slices bigger or smaller than a large slice?	$36\pi \div 6 = 6\pi \approx 18.85$ in², slightly smaller than a large slice.

Part 3: Design a Menu

#	Task	Answer
9	Choose three diameters and find each area.	Answers vary.
10	Set prices so that the price per square inch is about the same for all sizes. Show your work.	Answers vary.
11	Make a menu card with sizes, prices, and slices per pizza.	Answers vary.

Answer Key – CP1

Part 1: Value & Slices

#	Task	Answer
1	Compare one large with two mediums using exact areas and price per square inch.	Large $49\pi \approx 153.94$ in² for \$15 (≈ \$0.097 per in²); two mediums $50\pi \approx 157.08$ in² for \$18 (≈ \$0.115 per in²). The large is the better value.
2	A customer wants a 30 in² slice from a large pizza. What central angle should the cut have?	$\theta = 30 \div (49\pi) \times 360 \approx 70.2°$
3	A slice of a large pizza has a crust (arc) length of 7 in. Find its central angle and area.	$7 = (\theta \div 360) \times 14\pi \rightarrow \theta \approx 57.3°$; area = ½ · 7 · 7 = 24.5 in² (sector area = ½ × arc × radius)

Part 2: Cutting & Boxing

#	Task	Answer
4	A straight cut is made 3 in from the center of a large pizza. How long is the cut (the chord)?	Half-chord = $\sqrt{(7^2 - 3^2)} = \sqrt{40}$; chord = $2\sqrt{40} = 4\sqrt{10} \approx 12.65$ in
5	Three points on the crust form a triangle. One angle of the triangle intercepts an arc of 100°. What is that angle?	½ × 100° = 50°
6	A large pizza fits exactly in a square box. Each side of the box is tangent to the crust. How long is each side? Explain using tangents.	14 in, because the distance between parallel tangent lines equals the diameter.
7	The delivery zone is a circle centered at the shop (0, 0) with radius 5 miles. Write its equation. Is a house at (3, 4.5) in the zone?	$x^2 + y^2 = 25$. $3^2 + 4.5^2 = 29.25 > 25$, so no.

Part 3: Design a Menu

#	Task	Answer
8	Choose sizes and prices with a nearly equal price per square inch.	Answers vary.
9	Decide how many slices each size gets so that every slice is between 15 and 25 in². Show sector areas.	Answers vary.
10	Write the equation of your delivery zone and test two addresses.	Answers vary.

Answer Key – Honors

Part 1: Value & Shapes

#	Task	Answer
1	Show that the ratio of the areas of two pizzas equals the square of the ratio of their diameters. Use the large and medium.	$49\pi \div 25\pi = 49 \div 25 = (14 \div 10)^2$
2	A slice is cut along a straight chord (not through the center) that spans a 90° arc of a large pizza. Find the area of this segment.	Sector $12.25\pi \approx 38.48$ minus triangle $\frac{1}{2} \cdot 7 \cdot 7 = 24.5$: segment ≈ 13.98 in^2
3	Find the diameter of a single pizza with the same area as two mediums.	$\pi r^2 = 50\pi \rightarrow r = \sqrt{50} \approx 7.07$ in; diameter ≈ 14.14 in

Part 2: Circle Theorems

#	Task	Answer
4	Two cuts (chords) cross inside a pizza. One is divided into 4 in and 6 in; the other has a 3-in piece. Find the other piece.	$4 \cdot 6 = 3x \rightarrow x = 8$ in
5	From a point outside the pizza, a tangent segment is 6 in long. A secant from the same point passes through 4 in outside the pizza before reaching…	$6^2 = 4(4 + x) \rightarrow 36 = 16 + 4x \rightarrow x = 5$ in
6	The delivery zone is $x^2 + y^2 - 6x + 4y - 12 = 0$. Write it in standard form and find the center and radius.	$(x - 3)^2 + (y + 2)^2 = 25$; center $(3, -2)$, radius 5
7	Is the address (7, 2) inside the zone? What about (1, 1)?	(7, 2): $16 + 16 = 32 > 25$, outside. (1, 1): $4 + 9 = 13 < 25$, inside.

Part 3: Design a Pricing Model

#	Task	Answer
8	Write $P(d) = a + b \cdot (\text{area})$ with a base cost a and a price per square inch b. Choose a and b and justify them.	Answers vary.
9	Show that with a base cost, larger pizzas are a better value per square inch.	$P \div \text{area} = a \div \text{area} + b$, which decreases as the area increases.
10	Present a menu with three sizes and your formula.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Prices are fair (similar price per square inch) and every circle calculation is correct. Goes beyond the minimum with a creative, realistic choice.	Prices are fair (similar price per square inch) and every circle calculation is correct.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Menu proposal shows each size's area, price per square inch, slice sizes, and a pricing recommendation. Polished, organized, and visually engaging.	Menu proposal shows each size's area, price per square inch, slice sizes, and a pricing recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.